Mind Dreamer: Untethering Imagination via Active Causal Intervention on Latent Manifolds

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1. Abstract & Key Contributions

The Paradigm Shift:

Traditional latent imagination is trapped in an "information cage" of historical trajectories. We break this curse through **Active Causal Intervention**.

Core Concept:

Model-Based RL (e.g., DreamerV3) achieves sample efficiency via **latent imagination**—simulating futures within a learned world model anchored by the **Manifold Hypothesis**: observations reside on low-dimensional manifold $\mathcal{M} \subset \mathbb{R}^d$ where $d \ll \dim(\mathcal{X})$.

The Challenge — Historical Tethering:

Imagination must initialize from observed states ($s_0 \sim \mathcal{D}$), creating critical **Learning Asymmetry**:

- World model rapidly discovers global manifold structure $\mathcal{M}$ via dense self-supervision
- Policy optimization for sparse rewards remains trajectory-bound and slow
- Result: Imagination confined to previously traversed regions

Our Solution: Mind Dreamer Framework

We propose **Active Causal Intervention (ACI)** enabling non-continuous **latent jumps** to epistemic blind spots that are physically plausible yet cognitively challenging:

$$s_0 \sim p_{gen}(\cdot) \quad \text{(adversarial generator)}$$

Core Achievements

- ✓ **1.67× average speedup** over DreamerV3 on DeepMind Control Suite
- ✓ Up to **8.8× acceleration** in sparse-reward bottleneck tasks
- ✓ Formal proof of **Quadratic Discount** γ^2 for stable uncertainty propagation across discontinuities

2. Motivation: The "Historical Tethering" Bottleneck

Learning Asymmetry in MBRL:

The world model's manifold discovery vastly outpaces the policy's sparse-reward optimization:

> *"The world model sees through the world, but the agent cannot escape the 'starting village' due to Markovian continuity."*

The Trap of Markovian Continuity:

Standard MBRL initializes imagination from replay buffer: $s_0 \sim \mathcal{D}$

- Agent's imagination is **deadlocked** around historical trajectories
- Facing OOD high-reward regions, forced into inefficient random walk
- Cannot leverage world model's structural knowledge to bridge unseen areas

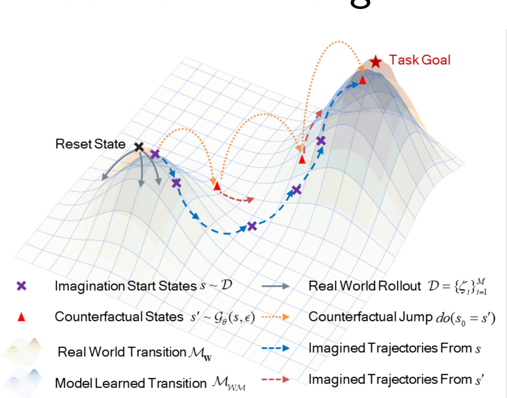

Figure 1: Manifold Intervention Visualization. Left (Blue): Standard MBRL tethered to historical buffer—imagination cannot escape bottlenecks. **Right (Red/Orange):** Mind Dreamer performs Latent Jumps via adversarial generator $\mathcal{G}$, establishing strategic anchors at manifold frontiers to bridge OOD regions.

3. Problem Formulation: Discovery vs. Optimization

4. Active Causal Intervention via Adversarial Synthesis

Core Idea:

Use an "adversarial generator" that specializes in teleporting the agent to the most challenging and worthwhile latent blind spots.

Adversarial Generator $\mathcal{G}_\theta$:

Instead of sampling from buffer, generate interventional initial states:

$$s_0 \sim p_{gen}(\cdot), \quad s' = \mathcal{G}_\theta(s, \epsilon)$$

Generator Objective balances three objectives:

1. **Epistemic Value** (unknownness): Identify regions where world model's tangent space is ill-defined
2. **Pragmatic Value** (potential high returns): Target states with high value function potential
3. **Manifold Self-Conformality** $\mathcal{L}_{mf}$: Ensure generated states are physically valid and reachable

$$\max_\theta \mathbb{E}\left[\eta V_{RVF}(s, s') + \beta V_{RUF}(s, s') - \lambda \mathcal{L}_{mf}(s')\right]$$

Manifold Constraints $\mathcal{L}_{mf}$:

- **Dynamics Coherence**: Penalize transition entropy $\mathcal{H}(p_\psi(\cdot|s', a))$ to prevent exploiting cracks in dynamics
- **Cycle-Consistency**: Enforce $D_{KL}[\mathrm{Enc}(\mathrm{Dec}(s'))\|s']$ to eliminate spatial hallucinations

Result: **43.5× reduction** in jump consistency error.

5. The Mind Dreamer System Pipeline

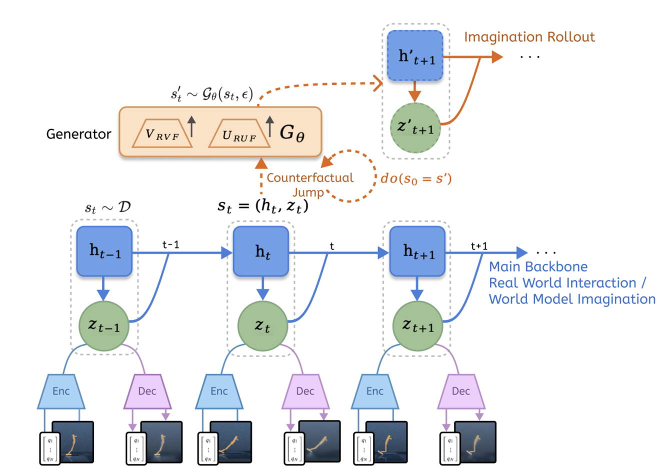

Figure 2: Mind Dreamer Architecture. Two core loops: (1) **Adversarial Synthesis Loop**: Generator $\mathcal{G}_\theta$ creates interventional anchors at manifold frontiers using RVF/RUF guidance. (2) **Relay Guidance Loop**: Policy performs dual-track imagination from both historical buffer and synthesized anchors, accelerating convergence via Bellman-style backups across spatial ruptures.

6. Bridging Discontinuity: Relay Potential Fields

The Credit Assignment Paradox:

When agent jumps to synthetic anchor s', a **spatial rupture** exists between current state s and target s'. Standard temporal difference methods fail across non-continuous trajectories.

We propose two recursive functionals treating anchors as **interventional intermediary states**:

A. Pragmatic Proxy: Relay Value Function (RVF)

Treats s' as gateway (not terminal destination):

$$V_{RVF}(s, s') \triangleq \max_\pi \mathbb{E}\left[\sum_{t=0}^{k-1} \gamma^t r_t + \gamma^k V_\phi(s')\right]$$

Intuition: Evaluates whether state s can be attracted and connected to the "virtual strong magnet" s' within k steps, opening strategic pathways.

Identifies **Pragmatic Gateways**: strategically positioned states unlocking high-density reward regions.

8. Theoretical Guarantees

Theorem 1: Contraction Mapping

Relay potential operators $\mathcal{T}_V$ and $\mathcal{T}_U$ are contraction mappings under L_∞ norm with factors γ and γ^2, respectively. This guarantees unique fixed points V^*_{RVF} and V^*_{RUF} representing optimal potential landscapes.

Theorem 2: Variance Reduction

Active Causal Intervention is mathematically equivalent to constructing an **Optimal Importance Sampler** for manifold boundary refinement:

$$q^*(s) \propto \rho(s) \|\nabla G(s)\|_2$$

Minimizing R-EFE aligns generator distribution with optimal proposal.

Proposition 1: Fisher Information Catalyst

RUF approximates trace of FIM:

$$\mathbb{E}[\mathcal{I}] \approx \frac{1}{2}\mathrm{Tr}(\mathcal{F}(\theta)\Sigma_\theta)$$

Maximizing RUF synthesizes anchors at high-curvature regions for maximal parameter refinement (**Manifold Repair**).

Theorem 3: Topological Acceleration

Traditional continuous exploration bounded by low conductance Φ. MD introduces discontinuous jumps, reducing coverage time complexity:

$$\nu = \frac{\tau_{traj}}{\tau_{MD}} \approx 1 + \chi^2(q^* \| q_{traj}) \propto \Phi^{-2}$$

From $\mathcal{O}(\mathrm{poly}(\Phi^{-1}))$ to $\mathcal{O}(\log|\mathcal{M}|)$.

Theorem 4: Hallucination Error Bound

Under L-Lipschitz value field:

$$\epsilon_V = \|V_{RVF}(s') - V^*(s')\| \leq \frac{L \cdot \delta}{1 - \gamma^n}$$

where $\delta = \|s' - \mathrm{Proj}_\mathcal{M}(s')\|$ is manifold deviation.

9. Experimental Results: DMC Benchmarks

Sample Efficiency Dominance:

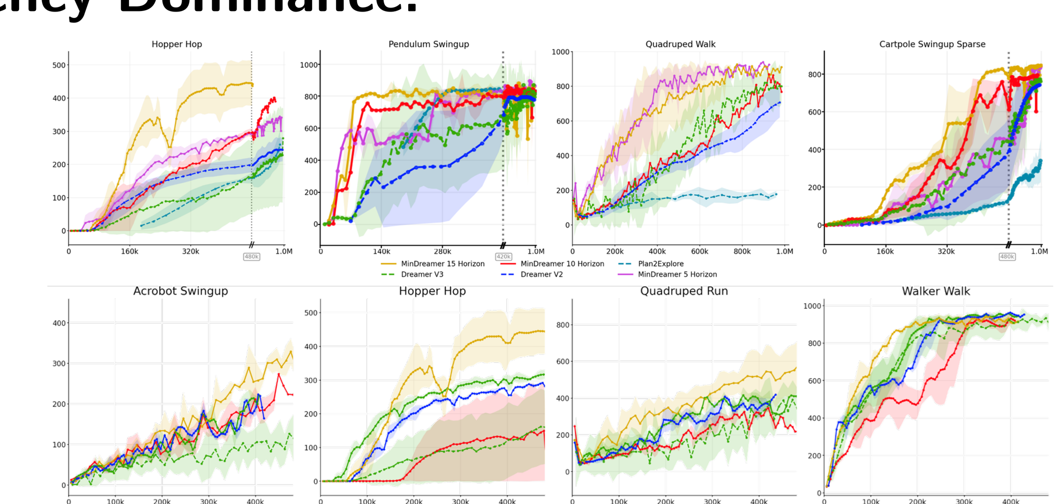

Figure 3: DeepMind Control Suite Results. Top: Performance curves demonstrating superior sample efficiency. Mind Dreamer (Gold) shows steeper convergence slope. **Bottom:** Ablation study validating necessity of each component. Shaded regions: std over 5 seeds.

Key Findings:

- ✓ **Average 1.67× speedup** over DreamerV3 across 20 DMC tasks
- ✓ Reaches 90% peak performance in **334.7k steps** vs. DV3's 557.6k steps
- ⋆ **Bottleneck tasks:** **>8.8× acceleration** (e.g., Pendulum Swingup)

Asymptotic Performance: